

Schnorr's Proof of Knowledge

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Protocols for Proof of knowledge

1 Schnorr's Dlog Proof of Knowledge

1.1 Protocol

- $g \rightarrow$ Group generator
- $p = \text{ord}(g)$

Prover ($w, x = g^w$)

- $r \xleftarrow{R} Z_p$
- $a = g^r$

$$z = r + c.w$$

$$\begin{array}{c} \longrightarrow \xrightarrow{a} \longrightarrow \\ \longleftarrow \xleftarrow{c} \longleftarrow \\ \longrightarrow \xrightarrow{z} \longrightarrow \end{array}$$

Verifier ($x = g^w$)

$$c \xleftarrow{R} Z_p$$

Check $g^z \stackrel{?}{=} a.x^c$

1.2 Properties

1.2.1 Knowledge Extractability/ Soundness

The extractor runs the protocol twice while a remains the same in both runs.

Prover*(malicious) ($w, x = g^w$)

$$\begin{array}{c} \longrightarrow \xrightarrow{a} \longrightarrow \\ \longleftarrow \xleftarrow{c^*} \longleftarrow \\ \longrightarrow \xrightarrow{z^*} \longrightarrow \end{array}$$

Extractor ($x = g^w$)

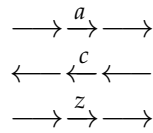
- $\frac{g^z}{g^{z^*}} = x^{c-c^*} = g^{w(c-c^*)}$
- $w = \frac{z-z^*}{c-c^*}$

1.2.2 Zero knowledge HVZK

HVZK because if V^* chooses c adaptively (not randomly), then Simulator is NOT Indistinguishable.

Simulator ($w, x = g^w$)

Verifier*(mal) ($x = g^w$)



- $z \xleftarrow{R} Z_p$
- $c \xleftarrow{R} Z_p$
- $a = \frac{g^z}{x^c}$

1.2.3 Completeness

It is trivial.